

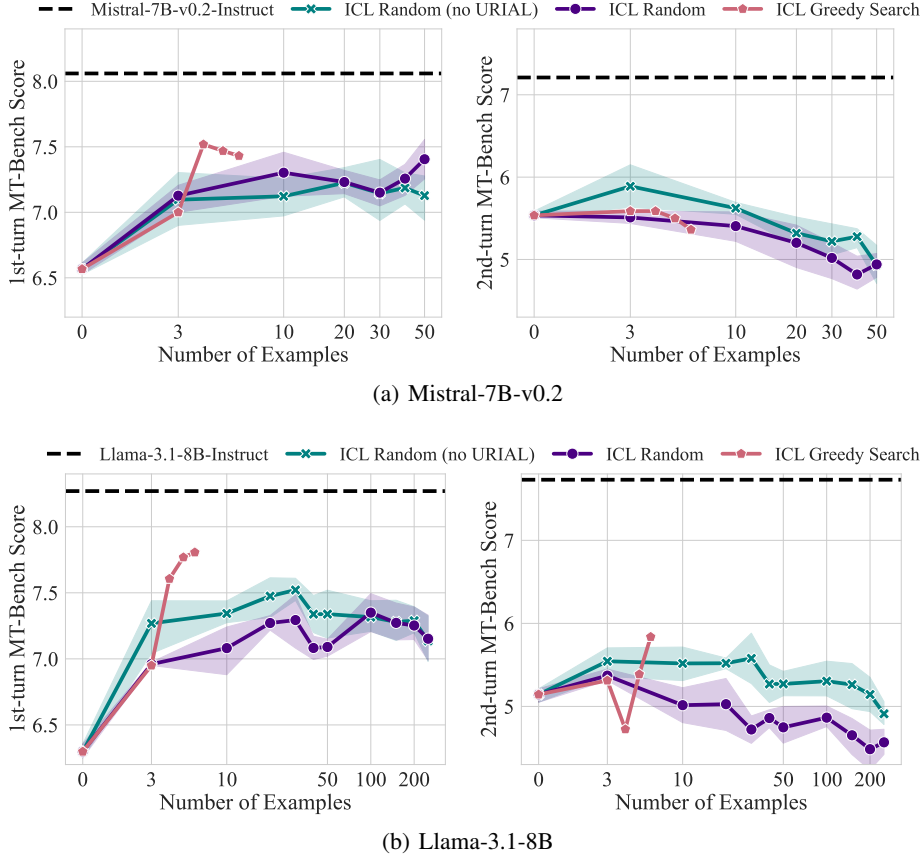


Figure 3: **Scaling the number of demonstrations for alignment with ICL on Mistral-7B-v0.2 and Llama-3.1-8B.** We measure the alignment performance of different settings using the MT-Bench score. ICL with more demonstrations quickly saturates and does not bridge the performance gap between the base model and its aligned counterpart. In particular, the ICL alignment performance of 3 random examples from the high-quality SkillMix dataset surpasses that of 3 examples from URIAL.

3.1 SCALING UP THE NUMBER OF IN-CONTEXT DEMONSTRATIONS

Given the success of many-shot ICL (Agarwal et al., 2024), we test the effect of increasing the number of in-context demonstrations. We sample random examples from the SkillMix dataset, since it contains high-quality question-answer pairs. We consider two scenarios: (1) all demonstrations are randomly chosen from SkillMix (the set of rules from URIAL is kept), and (2) we add the new demonstrations on top of the URIAL examples. In Fig. 3, we report the results on MT-Bench when varying the number of demonstrations. We repeat sampling with 5 random seeds and show mean and standard deviation over the corresponding results. We observe that for both base LLMs, the 1st-turn score increases with 3, 10 and 20 examples, but then plateaus without clear benefits from scaling up beyond 30 demonstrations (for Llama-3.1-8B the trend becomes even slightly negative with more than 100 examples). Thus, we find that adding up to 30 high-quality examples from SkillMix can improve alignment via ICL. This result contrasts with the findings of Lin et al. (2024), which showed that URIAL performed better with 3 demonstrations than with 8.

Next, we observe that in-context alignment does not improve the second-turn score. In fact, adding more examples continues to decrease performance, even falling below that of the base models. This behavior is likely due to the presence of only single-turn examples in the prompt, causing the LLM to respond in the same style without adapting to more complex conversations. Overall, these results show that, unlike in the setting of Agarwal et al. (2024); Bertsch et al. (2024), simply scaling the number of ICL examples is not sufficient to consistently improve the instruction following performance.

Finally, we notice that on Llama-3.1-8B, using three examples from SkillMix attain, on average, better performance than using the three examples of URIAL. Overall, there is no significant difference

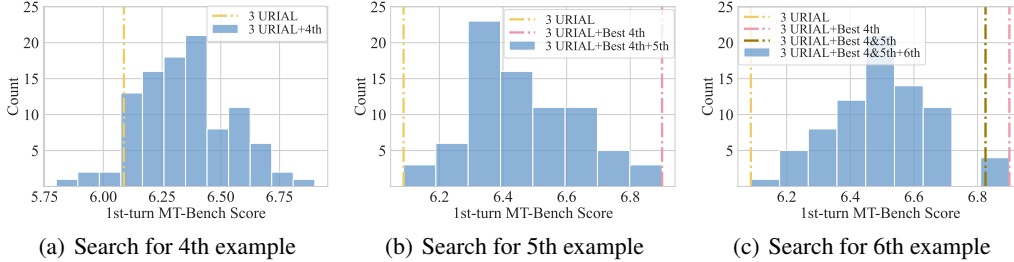


Figure 4: **The distribution of the 1st-turn MT-Bench score (GPT-4-Turbo as judge) on Llama-3.1-8B obtained by adding multiple instructions from SkillMix (Kaur et al., 2024) as a 4th (a), 5th (b), 6th (c) demonstration to URIAL.** The dashed lines of various colors refer to the 1st-turn MT-Bench score of the obtained searching results. A majority of the 4th examples contribute positively to the model’s instruction-following performance, but the improvement quickly diminishes when running the greedy search for 5th and 6th demonstrations.

Table 4: **Improved in-context (IC) alignment by adding to URIAL prompt the demonstrations found via greedy search.** We find the optimal IC prompts in an incremental way by selecting query-response pairs from the high-quality SkillMix dataset. We report the 1st-turn score on MT-Bench and the length-controlled (LC) win rates on AlpacaEval 2.0 when using different IC prompts.

Model	Mistral-7B-v0.2		Llama-3.1-8B	
	MT-Bench (1st)	AlpacaEval 2.0	MT-Bench (1st)	AlpacaEval 2.0
URIAL (3 examples)	7.00	8.22	6.95	7.28
URIAL + greedy search (1 ex.)	7.52	7.53	7.61	8.61
URIAL + greedy search (2 ex.)	7.47	7.78	7.77	8.16
URIAL + greedy search (3 ex.)	7.43	8.55	7.81	8.19

in the scaling behavior between using or not using the URIAL demonstrations. Thus, we conclude that the success of in-context alignment is only partially dependent on the demonstrations themselves, provided they are of sufficient quality (see also Sec. 4). We also include the results of scaling experiments (Sec. E.1) conducted on AlpacaEval 2.0, which consists of 805 examples.

3.2 GREEDY SEARCH FOR EFFECTIVE DEMONSTRATIONS

Given the limited success of adding random demonstrations to URIAL, we try to greedily maximize the MT-Bench score by testing 100 high quality instructions from SkillMix as the 4th additional example to URIAL. For each resulting ICL prompt, we compute the MT-Bench score with GPT-4-Turbo as judge instead of GPT-4 (used in MT-Bench) as the former is faster, cheaper, and helps mitigate potential overfitting to the benchmark score. We then repeat this procedure sequentially to find a 5th and a 6th demonstration, restricting the search space at each round to only instructions leading to a high enough MT-Bench score to reduce the computational cost (see details in App. A.3).

We add the results of the greedy search to the plots in Fig. 3, computing the true MT-Bench (i.e., with GPT-4 as judge). For both base models, the 4th example found by greedy search is sufficient to match the best 1st-turn score achieved by scaling the in-context examples with random demonstrations (see Sec. 3.1). For Llama-3.1-8B, the 5th and 6th demonstrations further improve the score, while they are not helpful for Mistral-7B-v0.2. In Table 4, we provide the details of these results: the 4th ICL example yields a significant improvement over URIAL, e.g., from 6.95 to 7.61 for Llama-3.1-8B, with only a further +0.20 given by the 5th and 6th. Overall, this evaluation further indicates that the improvement in instruction following one can achieve with in-context alignment quickly saturates when increasing the number of examples. In Fig. 4, we show the distribution of the scores (with GPT-4-Turbo as judge) at step of the greedy search when using Llama-3.1-8B as base model (the analog for Mistral-7B-v0.2 in App. E.2). Most demonstrations improve the score when added as the 4th example on top of URIAL (as shown in the first plot), but no further progress is observed with additional demonstrations.

We further test the IC prompts found by the greedy search on the AlpacaEval 2.0 (Li et al., 2023a) benchmark, where we measure length-controlled win rate. As shown in Table 4, using the examples

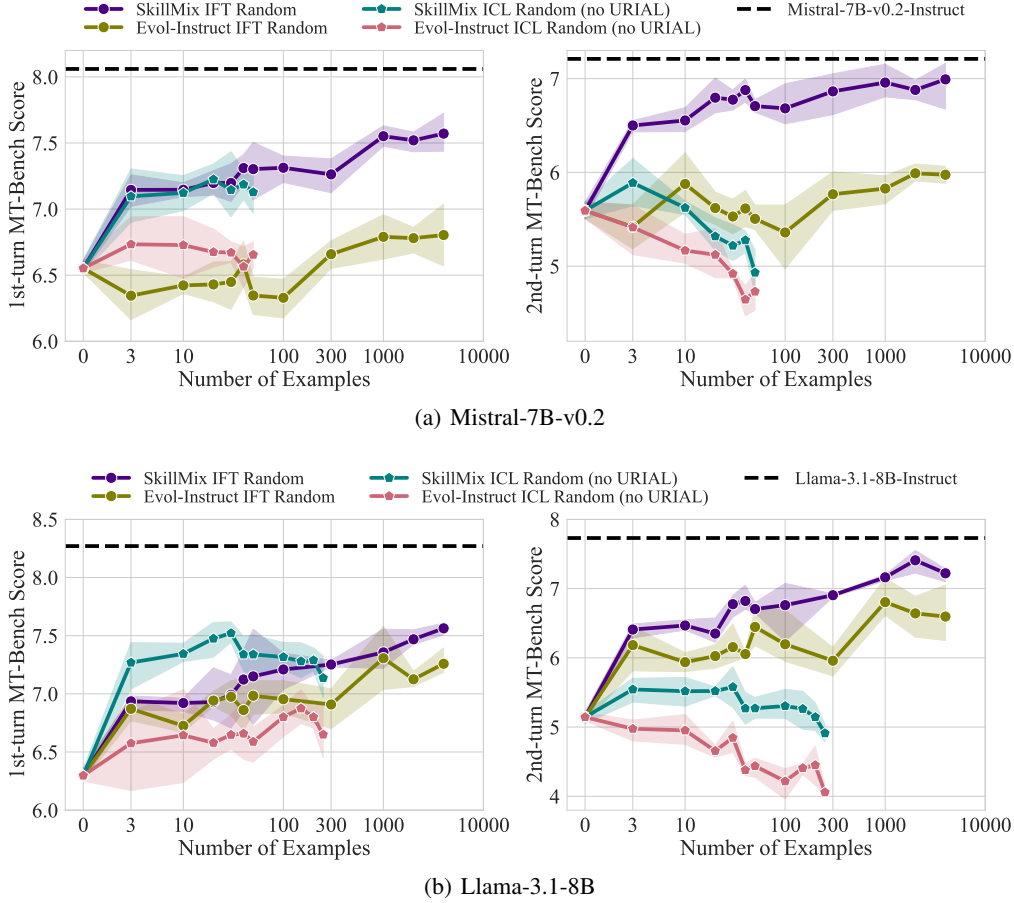


Figure 5: **Comparison of ICL vs IFT for alignment in the low data regime.** We measure the alignment performance of different settings for Mistral-7B-v0.2 and Llama-3.1-8B using the MT-Bench score. IFT with more demonstrations keeps improving the alignment performance, almost bridging the gap between the base model and its aligned counterpart. IFT-aligned models perform well on multi-turn conversations, unlike with ICL. Finally, data quality has significant impact on both IFT and ICL: the higher-quality SkillMix leads to better performance than Evol-Instruct.

given by greedy search leads to better results than plain URIAL on AlpacaEval 2.0, without directly optimizing for it. This improvement indicates that the found IC prompt does not completely overfit to the MT-Bench score, although the gains are less consistent than on MT-Bench. For Mistral-7B-v0.2, URIAL with 3 greedy search examples is the only configuration that outperforms URIAL, while for Llama-3.1-8B just one additional demonstration yields the best win rate, with a significant improvement over URIAL (7.28 vs 8.61).

4 A COMPARISON OF ICL VS IFT FOR INSTRUCTION FOLLOWING

The results from the previous sections strongly suggest that in-context alignment might not be as competitive as sophisticated, computationally heavy alignment techniques. However, it is unclear how it compares to more lightweight approaches, such as IFT, when applied to a small number of examples. In the low-data regime, the choice between ICL and IFT can be viewed as a decision between allocating resources for fine-tuning (which permanently modifies the model’s weights) or increasing inference time by adding the in-context prompt each time.

Setup. We test IFT on the same number of instructions from SkillMix as used in the many-shot ICL experiments in Sec. 3, ranging from 3 to 250 (the maximum allowed by the context length of the base models). Moreover, we scale the dataset up to 4k, that is the entire SkillMix. Each training run is repeated over multiple seeds, resulting in different training datasets (more details in App. A.4).